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|---|---|---|---|
| 0 | 4 | . | 1 |
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Describe how the process of meiosis results in haploid cells.

Do **not** include descriptions of how genetic variation is produced in meiosis.

**[4 marks]**

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**Question 4 continues on the next page**

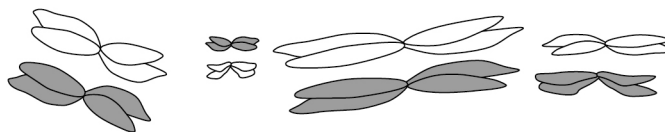
**Turn over ►**



0 4 . 2

**Figure 2** shows the arrangement of chromosomes in a cell during the first meiotic division.

**Figure 2**



A scientist observed 300 cells. All of the cells were at exactly the same stage of meiosis as the cell shown in **Figure 2**.

Use your knowledge of the independent segregation of homologous chromosomes to calculate how many of these cells are expected to have an **identical arrangement** of chromosomes to those shown in **Figure 2**. Assume no crossing over occurs.

**[2 marks]**

Answer \_\_\_\_\_

0 4 . 3

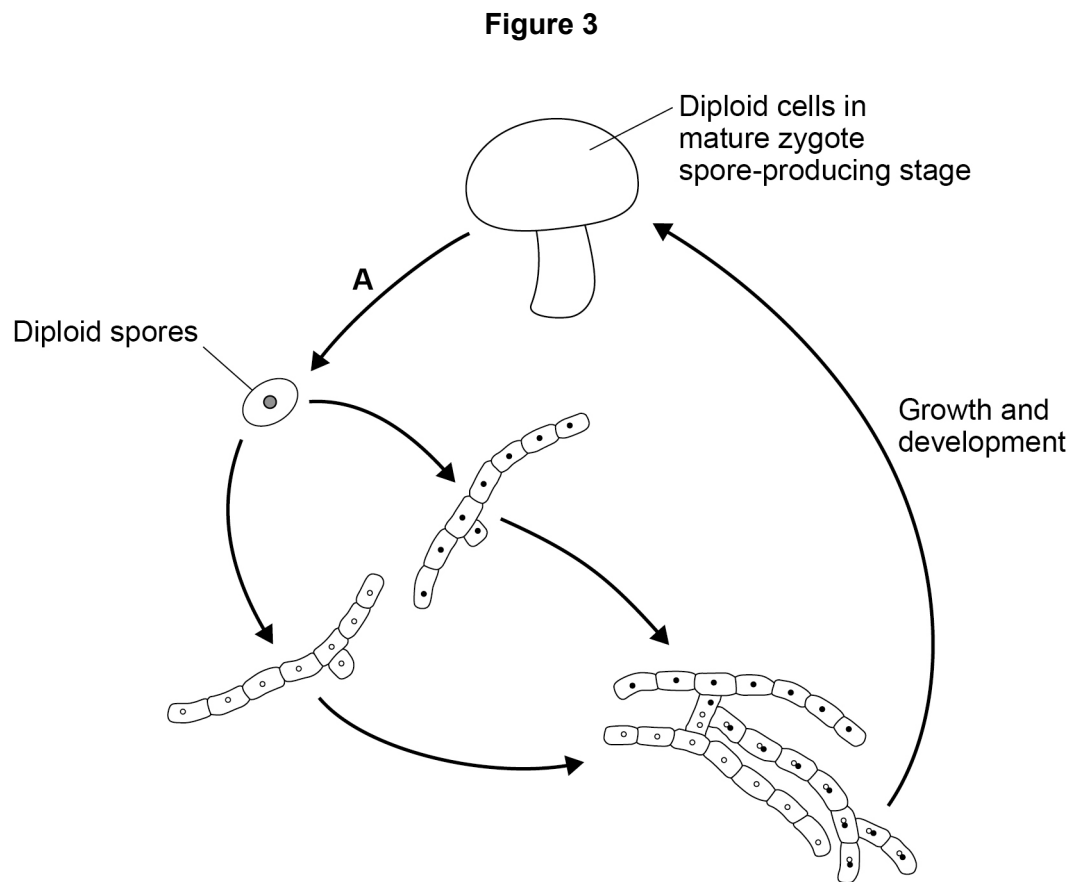
Draw a diagram to show the chromosomes in one gamete produced by meiosis from the cell shown in **Figure 2**.

**[1 mark]**



0 4 . 4

**Figure 3** shows the life cycle of a fungus. The life cycle includes sexual reproduction.



What is the name of the process shown by arrow **A** in **Figure 3**?

Tick (✓) **one** box.

[1 mark]

Binary fission

☐

Fertilisation

☐

Meiosis

☐

Mitosis

☐

**Turn over for the next question**

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